Tom de Groot

AI Engineer | Scientific AI, Evaluation & Data

Groningen, Netherlands | +31 6 314 90 416 | tomthegreatest04@gmail.com
github.com/Tomodovodoo | linkedin.com/in/tomodovodoo | x.com/Tomodovodoo

PROFILE

AI engineer and chemistry-trained researcher building evaluation, data, and agentic systems. Production work spans prediction algorithms, AI automation, AWS/container deployment, and full-stack applications. Research includes GNN-based NMR prediction, molecular data cleaning, HPC training, and controlled model evaluation. Open-source work combines simulator-first RL infrastructure with Lean-verified evaluation of LLM-generated mathematics.

EXPERIENCE

AI Engineer - SoundImports
Present

- Develop prediction algorithms and AI automations; deploy containerized services on AWS and build full-stack applications.
- Work across data, model, API, and deployment layers to turn operational workflows into reliable production systems.

Chemistry Research - University of Groningen, Pollice Group
2024 - 2025

- **BSc thesis:** Evaluating and Optimizing NMR Prediction with Graph Neural Networks.
- Curated molecular/NMR data and ran controlled GNN experiments on HPC, comparing architectures and model outcomes through a hypothesis-test-measure-iterate workflow.

Regional Director / QA - Teamspot
2022 - 2024

- Coordinated regional school projects, trained users, and bug-tested web, app, API, and integration workflows in Jira.

SELECTED EVALUATION, DATA & AI WORK

Bazaar_ML - Simulator, Certification & RL Environment
2026 - Present

- Built a simulator-first ML stack that ingests versioned public data, compiles heterogeneous mechanics into executable records, emits deterministic traces, and exposes agent-training hooks.
- Designed schema-frozen certification and provenance. The current committed matrix contains **49,248 cases**, **99.85%** executable sampling-pool coverage, content/parser hashes, explicit unsupported primitives, and zero recorded execution failures.

LLM-Aided Formal Mathematics - Lean 4
2025 - Present

- Orchestrated multiple agents to solve FirstProof Problem 9 informally and formalize it in Lean; the repository pins a reproducible toolchain and contains the only solved problem in that workspace.
- Co-authored multiple merged PNT+ proofs and use Lean compilation, tests, and adversarial review as hard evaluations for model output.

BaanSoliciteren - Agentic Data Pipeline
2025 - 2026

- Built a Python/Gmail MCP workflow to ingest vacancy pages and email, normalize them into JSON, score fit, track state, and archive outcomes with a reproducible audit trail.

EDUCATION & RECOGNITION

BSc Chemistry, University of Groningen - Chemistry of Life track, 2021 - 2025 | **Harmonic Rising Mathematician Award**, 2026 | **Chess.com Member of the Year**, 2021

TECHNICAL SKILLS

Evaluation & data: benchmark/certification design, data quality and provenance, dataset versioning, deterministic replay, failure analysis, scientific experimentation
AI & engineering: Python, PyTorch, TensorFlow, scikit-learn, pandas, NumPy, SQL, RDKit, JavaScript, APIs, AWS, Docker, Linux/HPC, Git
Formal methods: Lean 4, mathlib, Lake, CI | **Science:** analytical chemistry, biochemistry, NMR, BSL-2, molecular data | **Languages:** Dutch (native), English (fluent)